CE5

Project Introduction

CE5.1

Project Name	ENSE3020 Midterm Project: Workstation Monitor
Project Duration	07/2020 to 11/2020
Project Organization	The University of Western Australia
My Title	Electronics Engineer

CE5.2

Designed to monitor a workstation containing multiple storage drives, this system specifically checks if the internal temperature reaches a maximum threshold and verifies the normal operation of its three cooling fans. Monitoring these parameters is crucial, as overheating and fan failures are common causes of workstation malfunction or damage. Our design is structured into two modules:

(1) Internal Monitoring (our design): This module is positioned inside the workstation. Its primary function is to collect real-time temperature data and fan status information, achieved by an ATMEGA328P microcontroller.

(2) External Display (bought): Located outside the workstation, this module is a mini-LED display. Its purpose is to provide clear visual alerts if a fault is detected from the workstation.

The project team achieved the desired monitoring functionality in a very low-cost manner by avoiding the use of commercial Arduino microcontroller boards. This low-cost approach is particularly useful for assisting with the maintenance of lower-cost workstations, providing an economical and efficient monitoring solution.

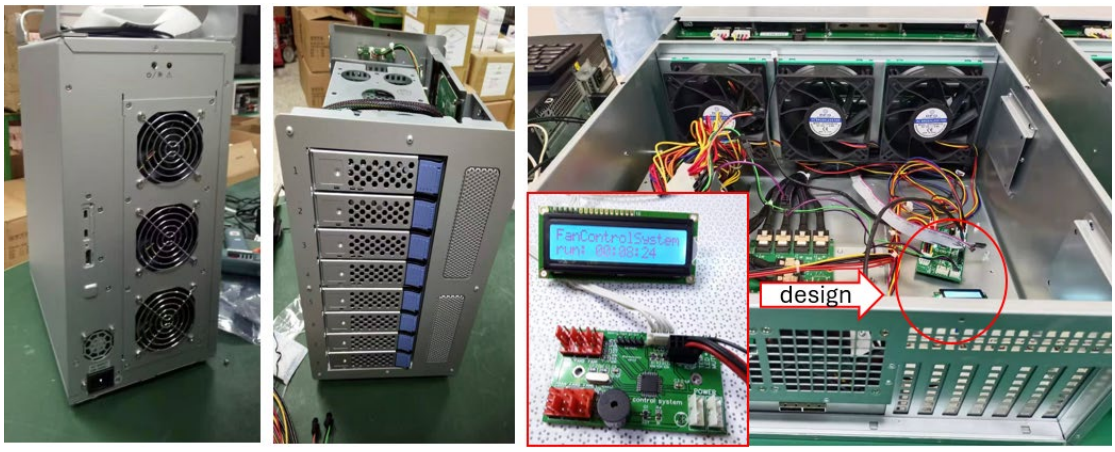
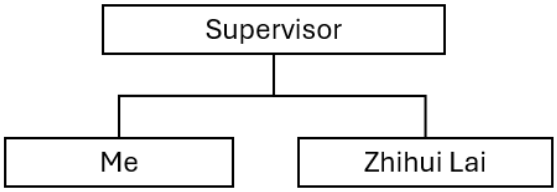


Figure 1: Overview of our Workstation Monitor. From Left to Right: (1) workstation back (2) workstation front view (3) workstation insider and our monitor installed.

Project Background

CE5.3

This design was undertaken as a two-person mid-term project for ENSC3020. Together with my teammate Zhihui Lai, we developed a comprehensive project plan, clearly defined our respective roles and established effective cooperation.



CE5.4

The main objective of this project is to design a workstation monitor (fans & temperature).

CE5.5

The main tasks of this project included software design, hardware design, and optimization based on the budget. Zhihui and I collaborated by dividing responsibilities. **My focus was the hardware design, while Zhihui was taking more responsive of coding.** Starting with the commercial Arduino boards, I progressively optimized the module's electronic design, moving from the higher-priced (1) Arduino Uno to the (2) Arduino Nano, and finally implementing it using the (3) ATMEGA328P. This optimization resulted in a display module with a smaller size and a significantly lower cost.

Project Activities

CE5.6 Overview

- I designed a monitoring system to track the internal temperature and cooling fan status of the workstation in real time, aiming to respond to potential abnormal conditions. Since each fan driver already provided temperature and speed feedback, I decided not to install additional standalone sensors and instead read both types of data directly from the fan drivers. To ensure the system could accurately acquire and process this information, I designed the appropriate data reading ports, integrated the data streams from multiple fans, and wrote custom code in C to extract key information and drive the LCD for status display and fault alerts. To implement an effective warning mechanism, I set logical conditions so that when a fan stops rotating, a “fan fault” message appears on the LCD, and when the temperature exceeds the threshold, an “overheat” warning is displayed—thereby enhancing the safety and reliability of system operation.
- I selected the ATMEGA328P microcontroller as the central processing unit to execute control logic for data acquisition, signal conversion, and fault detection. To ensure efficient interaction with specific hardware components, I configured its input pins to read digital pulse signals from the fan tachometers. In the program, I set fault detection conditions so the

microcontroller could control a mini-LED on the external display module based on the results, providing a clear visual alert. At the same time, I decided to power the microcontroller, all connected sensors, and the LED display module using a stable DC voltage source derived from within the workstation, ensuring voltage stability and reliable system response during operation.

CE5.7 Hardware Implementation

- During the circuit design phase, I used Fritzing software to draw the complete system schematic, enabling modular and visual layout management while ensuring clear logical connections between functional units. For the core components, I selected the ATMEGA328P microcontroller as the central processing unit, paired it with an LCD display for real-time output, and included resistors, capacitors, fan tachometers, and PC data readers to build a comprehensive data acquisition and display system. To improve flexibility during physical assembly, I reserved ample connection ports in the design, including multiple input and output interfaces, to accommodate component layout adjustments and potential future function expansions. I also paid close attention to the operability during soldering, arranging components to avoid wire crossings and enhance reliability and assembly efficiency. For specific interface configurations, I designated the input ports for receiving data from fan tachometers and temperature sensors, while the output ports were used to control the LCD display for showing system status and fault messages. I also added a dedicated power input port to ensure a stable voltage supply. In terms of user interaction, I designed two functional buttons—one for system reset, startup, and screen switching, and the other as a reserved interface for future use. To maintain stable logic levels for the buttons, I added pull-up resistors in the circuit to hold the pins high when not pressed and limit current when pressed to ground, thereby preventing damage to the control unit caused by short circuits or excessive current. While drawing the schematic, I repeatedly verified the logical connections between modules, ensured the accuracy of signal transmission directions, and analysed potential interference paths to optimize wiring and component layout, laying a solid technical foundation for subsequent PCB design and prototype development.

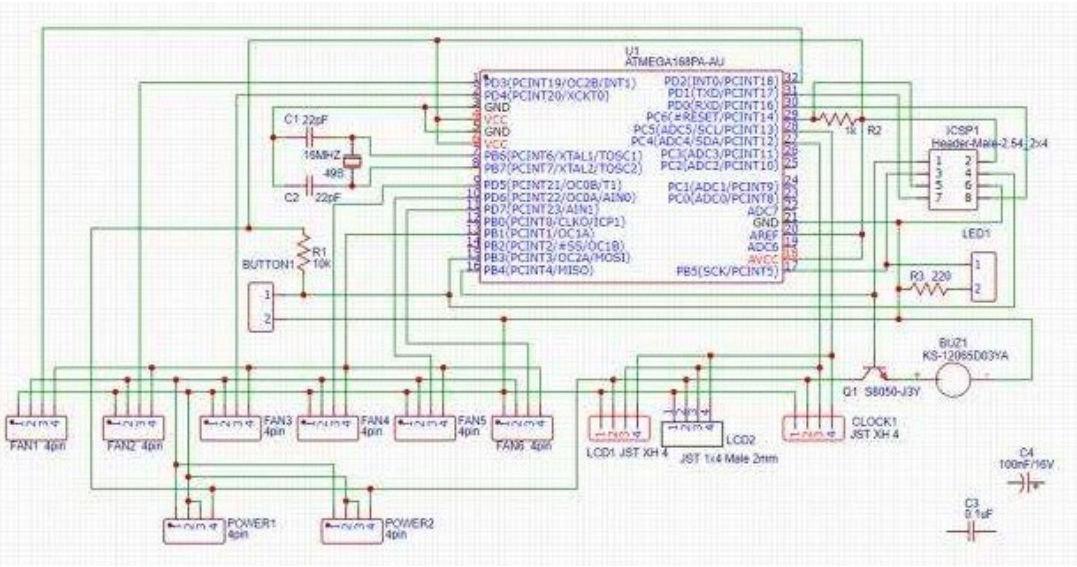


Figure 2: *Circuit Schematic of our Internal Monitor, created via the Windows App – Fritzing. This figure is low-quality preview-version saved from Chat History, as the original file was lost.*

- After completing the circuit design, I further developed the PCB layout based on the schematic and decided to submit the design files to a factory via an online platform for custom manufacturing, ensuring precision and production efficiency. To simplify the installation of the display module, I directly purchased the plug-and-play SMR1602-N mini-LCD module, avoiding potential compatibility issues during self-assembly. During the component soldering phase, I followed the design layout to solder each electronic component onto the PCB and connected the mini-LCD to the main board, thereby completing the assembly and physical realization of the entire hardware system.

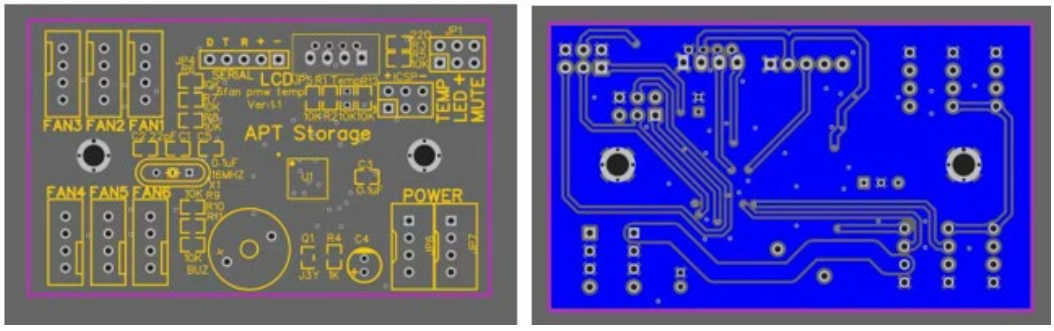


Figure 3: Optimised PCB Design of Our Circuit Sent for Manufacturing

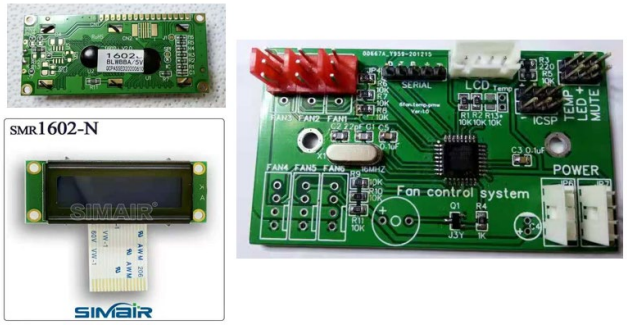
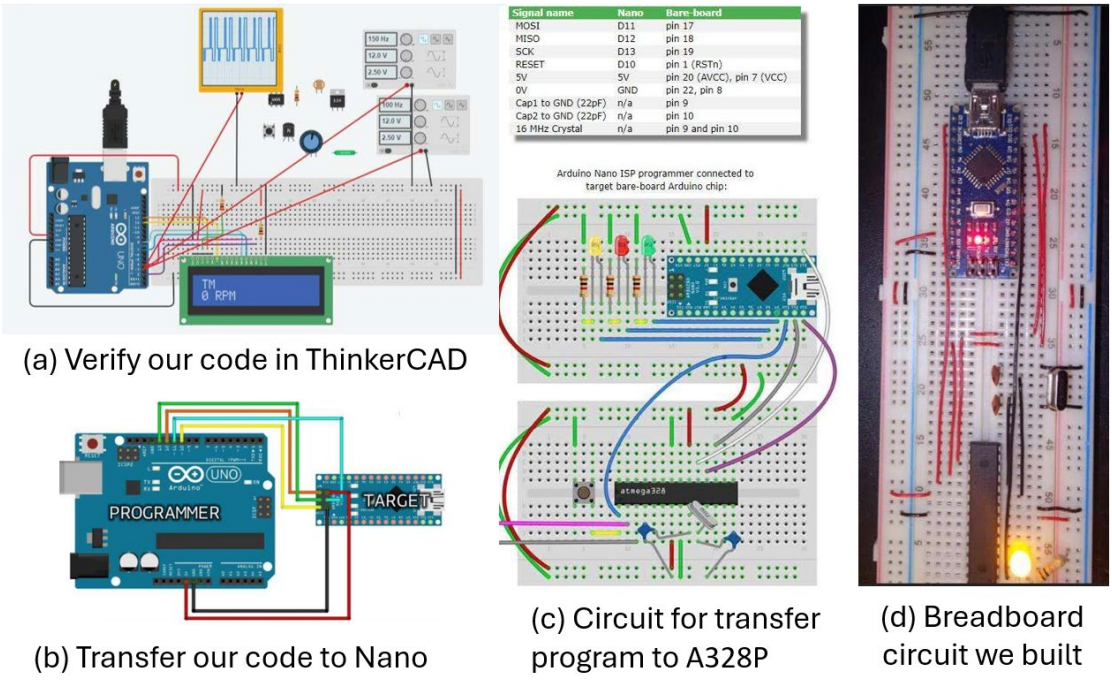


Figure 4*: The mini-LCD display we bought* ***(left)****, and PCB after soldered the components* ***(right)****.*

CE5.8 Hardware Optimization

- At the beginning, I used an Arduino UNO and C programming to implement the system's display and alert functionalities. Considering that our target users might be cost-sensitive, I later optimized the design by reducing the per-unit cost of the programmable control board. After a comparative analysis, I gradually shifted the solution from the $20 Arduino UNO to the more affordable $7.5 Arduino Nano, and finally decided to use the ATMEGA328P chip, which costs around $2, as the core controller. Despite the significant cost reduction, I evaluated the ATMEGA328P's performance under a 16 MHz external crystal and confirmed that its 32 KB Flash and 2 KB SRAM memory resources were sufficient to meet the project's requirements for data acquisition, processing logic, and display control. Therefore, I made the decision to switch to this more cost-effective control platform.
- After verifying the functionality of the code using TinkerCAD, a free online simulation platform (**Figure 7a**), I began the platform migration process. First, I uploaded the program that had been running on the Arduino UNO to the Arduino Nano to confirm its compatibility and stability on a lower-cost platform (**Figure 7b**). Next, I assembled the corresponding breadboard circuit and added an overvoltage protection circuit to prevent potential damage to the chip caused by abnormal voltage or wiring errors during programming (**Figures 7c and 7d**). In the implementation process, I configured the Arduino Nano as an In-System Programmer (ISP) and used it to successfully burn the program into the ATMEGA328P chip, thereby completing the transition from simulation verification to hardware deployment.



(a) Verify our code in ThinkerCAD

(b) Transfer our code to Nano

(c) Circuit for transfer program to A328P

(d) Breadboard circuit we built

Figure 5: Program Migration Workflow: From verification to final chip.

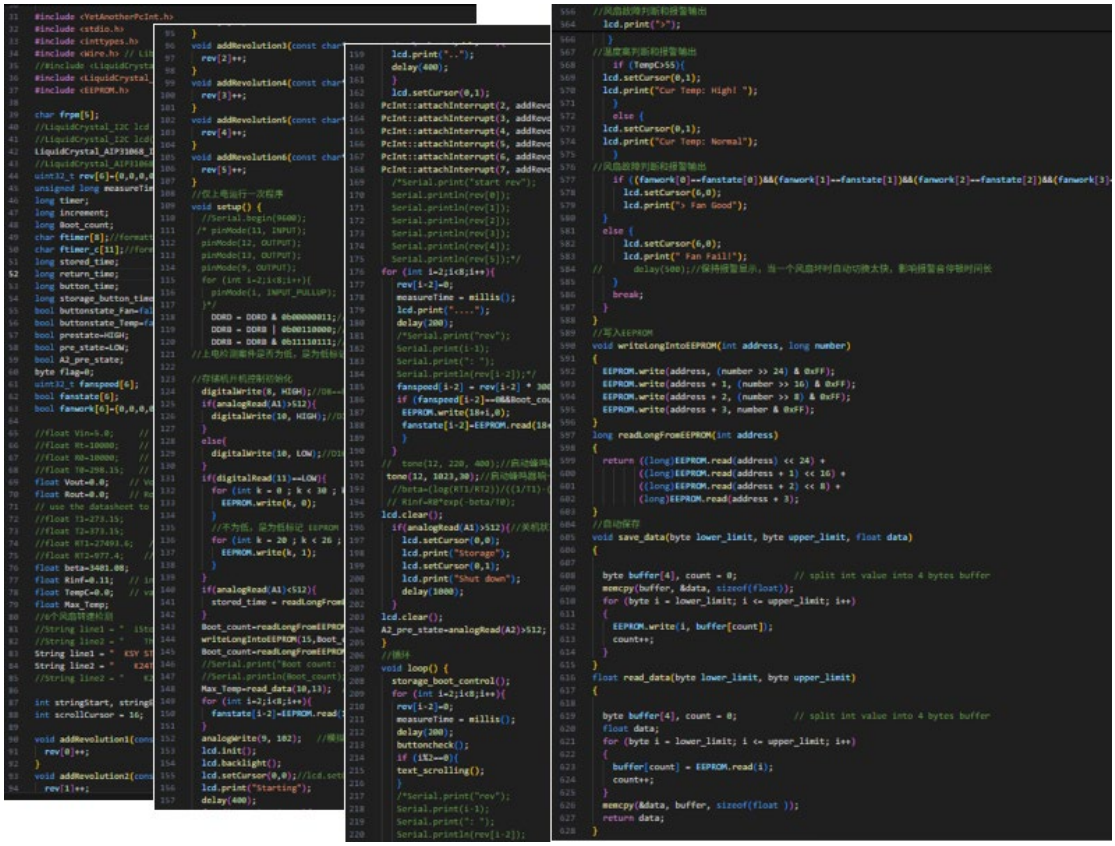


Figure 8: C language programming (by Zhihui Lai) for chip functionality.

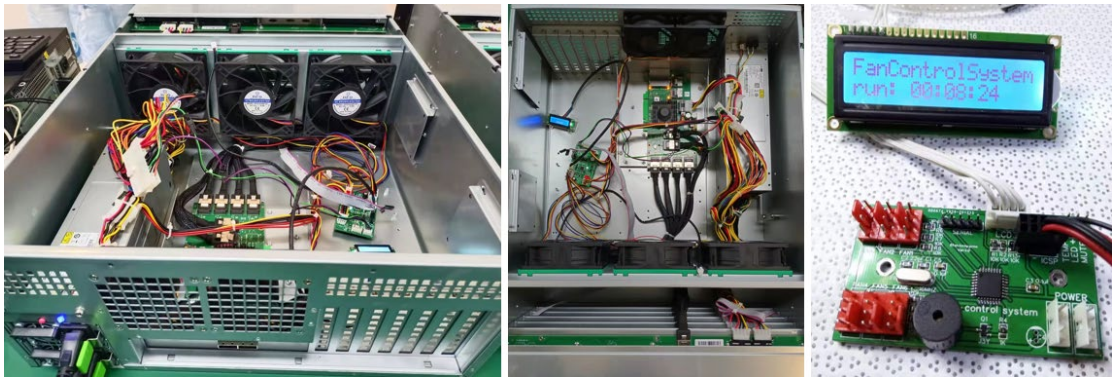


Figure 9: Final overview of our Workstation Monitor.

CE5.9

Due to the limited time available for this mid-term project, I decided not to include the design of a physical enclosure and instead focused on ensuring the correctness of the circuit design and program functionality. However, when evaluating the project's scalability, I recognized that a physical case would be necessary to transform the system into a practical, ready-to-use product. Considering the need to protect internal components and improve the overall appearance, I proposed using 3D printing as a feasible future solution for creating the enclosure, thus reserving a technical path for product completion without affecting the current development timeline.

Project Summary

CE5.11

Collaborating with Zhihui Lai (programming focus), my primary concentration was on circuit design throughout this project's phases. We diligently worked together, I paid strict attention to the project timeline, ensuring all processes were completed on time. Our research and discussions informed the design, leading to effective solutions like powering directly from the workstation. I am proud of the optimization achieved via careful chip selection and code migration. This reduced cost, fit our scenario “monitoring a cost-efficiency workstation”. Besides, this mid-project built a strong basis for our ENSC3020 final group project **(refer to my Career Episode 3).**